High Performance Computing

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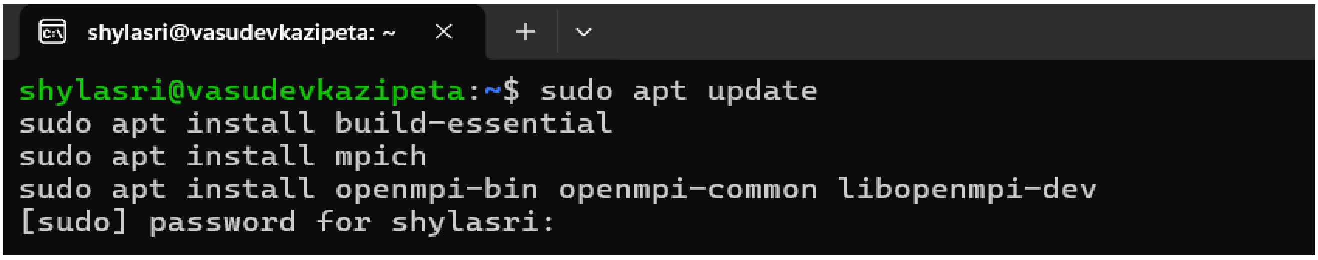
Batch-14

Week-10

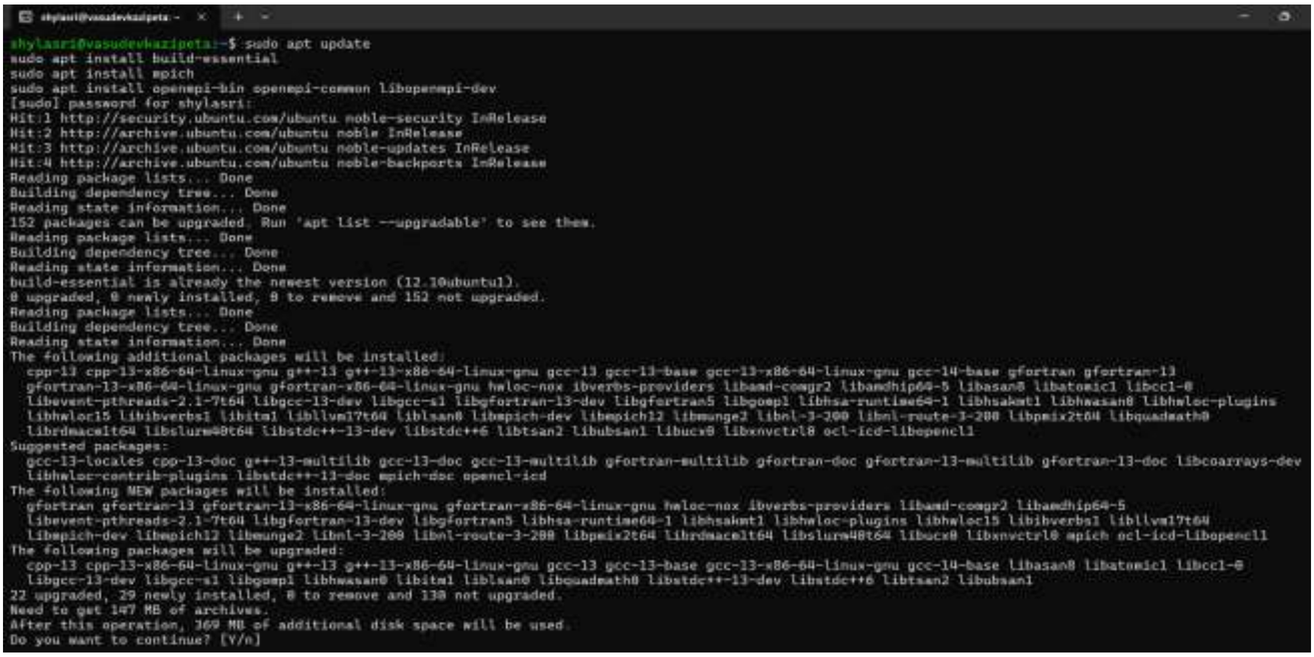
LAB Assignment-10 (25/02/2026)

Lab: MPI: Hello World and Send/Receive

Install Required Packages



```
shylasri@vasudevkazipeta:~$ sudo apt update
sudo apt install build-essential
sudo apt install mpich
sudo apt install openmpi-bin openmpi-common libopenmpi-dev
[sudo] password for shylasri:
```



```
shylasri@vasudevkazipeta:~$ sudo apt update
sudo apt install build-essential
sudo apt install mpich
sudo apt install openmpi-bin openmpi-common libopenmpi-dev
[sudo] password for shylasri:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
152 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 152 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cpp-13 cpp-13-x86-64-linux-gnu g++-13 g++-13-x86-64-linux-gnu gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu gcc-14-base gfortran gfortran-13
  gfortran-13-x86-64-linux-gnu gfortran-x86-64-linux-gnu hwloc-nox ibverbs-providers libamd-comgr2 libamdhip64-5 libasan8 libatomic1 libcc1-0
  libevent-pthreads-2.1-7t64 libgcc-13-dev libgcc-s1 libgfortran-13-dev libgfortran5 libgomp1 libhsa-runtime64-1 libhsakmt1 libhwasan0 libhwloc-plugins
  libhwloc15 libibverbs1 libitm1 libllvm17t64 liblsan0 libmpich-dev libmpich12 libmunge2 libnl-3-200 libnl-route-3-200 libpmix2t64 libquadmath0
  librdmacm1t64 libslurm40t64 libstdc++-13-dev libstdc++6 libtsan2 libubsan1 libucx0 libxnvctrl0 ocl-icd-libopencl1
Suggested packages:
  gcc-13-locales cpp-13-doc g++-13-multilib gcc-13-doc gcc-13-multilib gfortran-multilib gfortran-doc gfortran-13-multilib gfortran-13-doc libcoarrays-dev
  libhwloc-contrib-plugins libstdc++-13-doc mpich-doc opencl-icd
The following NEW packages will be installed:
  gfortran gfortran-13 gfortran-13-x86-64-linux-gnu gfortran-x86-64-linux-gnu hwloc-nox ibverbs-providers libamd-comgr2 libamdhip64-5
  libevent-pthreads-2.1-7t64 libgfortran-13-dev libgfortran5 libhsa-runtime64-1 libhsakmt1 libhwloc-plugins libhwloc15 libibverbs1 libllvm17t64
  libmpich-dev libmpich12 libmunge2 libnl-3-200 libnl-route-3-200 libpmix2t64 librdmacm1t64 libslurm40t64 libucx0 libxnvctrl0 mpich ocl-icd-libopencl1
The following packages will be upgraded:
  cpp-13 cpp-13-x86-64-linux-gnu g++-13 g++-13-x86-64-linux-gnu gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu gcc-14-base libasan8 libatomic1 libcc1-0
  libgcc-13-dev libgcc-s1 libgomp1 libhwasan0 libitm1 liblsan0 libquadmath0 libstdc++-13-dev libstdc++6 libtsan2 libubsan1
22 upgraded, 29 newly installed, 0 to remove and 130 not upgraded.
Need to get 147 MB of archives.
After this operation, 369 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

shylasri@vasudevkazipeta: ~

```
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libubsan1 amd64 14.2.0-4ubuntu2~24.04.1 [1184 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libstdc++-13-dev amd64 13.3.0-6ubuntu2~24.04.1 [2420 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13 amd64 13.3.0-6ubuntu2~24.04.1 [494 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 g++-13 amd64 13.3.0-6ubuntu2~24.04.1 [16.0 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 g++-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2~24.04.1 [12.2 MB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2~24.04.1 [21.1 MB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2~24.04.1 [10.7 MB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13 amd64 13.3.0-6ubuntu2~24.04.1 [1042 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-base amd64 13.3.0-6ubuntu2~24.04.1 [51.6 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgcc-13-dev amd64 13.3.0-6ubuntu2~24.04.1 [2681 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libtsan2 amd64 14.2.0-4ubuntu2~24.04.1 [2772 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-14-base amd64 14.2.0-4ubuntu2~24.04.1 [51.0 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libstdc++6 amd64 14.2.0-4ubuntu2~24.04.1 [792 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libquadmath0 amd64 14.2.0-4ubuntu2~24.04.1 [153 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 liblsan0 amd64 14.2.0-4ubuntu2~24.04.1 [1322 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libitm1 amd64 14.2.0-4ubuntu2~24.04.1 [29.7 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libhwasan0 amd64 14.2.0-4ubuntu2~24.04.1 [1641 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgomp1 amd64 14.2.0-4ubuntu2~24.04.1 [148 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcc1-0 amd64 14.2.0-4ubuntu2~24.04.1 [48.0 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libatomic1 amd64 14.2.0-4ubuntu2~24.04.1 [10.5 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libasan8 amd64 14.2.0-4ubuntu2~24.04.1 [3027 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgcc-s1 amd64 14.2.0-4ubuntu2~24.04.1 [78.4 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-3-200 amd64 3.7.0-0.3build1.1 [55.7 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-route-3-200 amd64 3.7.0-0.3build1.1 [189 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libibverbs1 amd64 50.0-2ubuntu0.2 [68.0 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ibverbs-providers amd64 50.0-2ubuntu0.2 [381 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgfortran5 amd64 14.2.0-4ubuntu2~24.04.1 [916 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgfortran-13-dev amd64 13.3.0-6ubuntu2~24.04.1 [928 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gfortran-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2~24.04.1 [11.4 MB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gfortran-13 amd64 13.3.0-6ubuntu2~24.04.1 [13.9 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble/main amd64 gfortran-x86-64-linux-gnu amd64 4:13.2.0-7ubuntu1 [1024 B]
Get:32 http://archive.ubuntu.com/ubuntu noble/main amd64 gfortran amd64 4:13.2.0-7ubuntu1 [1176 B]
Get:33 http://archive.ubuntu.com/ubuntu noble/main amd64 libllvm17t64 amd64 1:17.0.6-9ubuntu1 [26.2 MB]
Get:34 http://archive.ubuntu.com/ubuntu noble/universe amd64 libamd-comgr2 amd64 6.0+git20231212.4510c28+dfsg-3build2 [14.4 MB]
Get:35 http://archive.ubuntu.com/ubuntu noble/universe amd64 libhsakmt1 amd64 5.7.0-1build1 [62.9 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble/universe amd64 libhsa-runtime64-1 amd64 5.7.1-2build1 [491 kB]
Get:37 http://archive.ubuntu.com/ubuntu noble/universe amd64 libamdhip64-5 amd64 5.7.1-3 [9621 kB]
Get:38 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7982 B]
Get:39 http://archive.ubuntu.com/ubuntu noble/universe amd64 libhwloc15 amd64 2.10.0-1build1 [172 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 libmunge2 amd64 0.5.15-4ubuntu0.1 [14.8 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libxnvctrl0 amd64 510.47.03-0ubuntu4.24.04.1 [12.7 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble/universe amd64 ocl-icd-libopencl1 amd64 2.3.2-1build1 [38.5 kB]
Get:43 http://archive.ubuntu.com/ubuntu noble/universe amd64 libhwloc-plugins amd64 2.10.0-1build1 [15.7 kB]
Get:44 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpmix2t64 amd64 5.0.1-4.1build1 [697 kB]
Get:45 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 librdmacm1t64 amd64 50.0-2ubuntu0.2 [70.7 kB]
Get:46 http://archive.ubuntu.com/ubuntu noble/universe amd64 libslurm40t64 amd64 23.11.4-1.2ubuntu5 [553 kB]
Get:47 http://archive.ubuntu.com/ubuntu noble/universe amd64 libucx0 amd64 1.16.0+ds-5ubuntu1 [1140 kB]
Get:48 http://archive.ubuntu.com/ubuntu noble/universe amd64 hwloc-nox amd64 2.10.0-1build1 [221 kB]
Get:49 http://archive.ubuntu.com/ubuntu noble/universe amd64 libmpich12 amd64 4.2.0-5build3 [8247 kB]
Get:50 http://archive.ubuntu.com/ubuntu noble/universe amd64 mpich amd64 4.2.0-5build3 [202 kB]
```

shylasri@vasudevkazipeta: ~

```
Get:51 http://archive.ubuntu.com/ubuntu noble/universe amd64 libmpich-dev amd64 4.2.0-5build3 [10.2 MB]
Fetched 147 MB in 35s (4241 kB/s)
Extracting templates from packages: 100%
(Reading database ... 46772 files and directories currently installed.)
Preparing to unpack .../00-libubsan1_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libubsan1:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../01-libstdc++-13-dev_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking libstdc++-13-dev:amd64 (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../02-gcc-13_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking gcc-13 (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../03-g++-13_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking g++-13 (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../04-g++-13-x86-64-linux-gnu_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking g++-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../05-gcc-13-x86-64-linux-gnu_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking gcc-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../06-cpp-13-x86-64-linux-gnu_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking cpp-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../07-cpp-13_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking cpp-13 (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../08-gcc-13-base_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking gcc-13-base:amd64 (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../09-libgcc-13-dev_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking libgcc-13-dev:amd64 (13.3.0-6ubuntu2~24.04.1) over (13.3.0-6ubuntu2~24.04) ...
Preparing to unpack .../10-libtsan2_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libtsan2:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../11-gcc-14-base_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking gcc-14-base:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Setting up gcc-14-base:amd64 (14.2.0-4ubuntu2~24.04.1) ...
(Reading database ... 46772 files and directories currently installed.)
Preparing to unpack .../libstdc++6_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libstdc++6:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Setting up libstdc++6:amd64 (14.2.0-4ubuntu2~24.04.1) ...
(Reading database ... 46772 files and directories currently installed.)
Preparing to unpack .../0-libquadmath0_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libquadmath0:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../1-liblsan0_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking liblsan0:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../2-libitm1_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libitm1:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../3-libhwasan0_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libhwasan0:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../4-libgomp1_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libgomp1:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../5-libcc1-0_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libcc1-0:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../6-libatomic1_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libatomic1:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../7-libasan8_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libasan8:amd64 (14.2.0-4ubuntu2~24.04.1) over (14.2.0-4ubuntu2~24.04) ...
Preparing to unpack .../8-libgcc-s1_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
```

```
shylasri@vasudevkazipeta: ~
Setting up libgcc-s1:amd64 (14.2.0-4ubuntu2~24.04.1) ...
Selecting previously unselected package libnl-3-200:amd64.
(Reading database ... 46772 files and directories currently installed.)
Preparing to unpack .../00-libnl-3-200_3.7.0-0.3build1.1_amd64.deb ...
Unpacking libnl-3-200:amd64 (3.7.0-0.3build1.1) ...
Selecting previously unselected package libnl-route-3-200:amd64.
Preparing to unpack .../01-libnl-route-3-200_3.7.0-0.3build1.1_amd64.deb ...
Unpacking libnl-route-3-200:amd64 (3.7.0-0.3build1.1) ...
Selecting previously unselected package libibverbs1:amd64.
Preparing to unpack .../02-libibverbs1_50.0-2ubuntu0.2_amd64.deb ...
Unpacking libibverbs1:amd64 (50.0-2ubuntu0.2) ...
Selecting previously unselected package ibverbs-providers:amd64.
Preparing to unpack .../03-ibverbs-providers_50.0-2ubuntu0.2_amd64.deb ...
Unpacking ibverbs-providers:amd64 (50.0-2ubuntu0.2) ...
Selecting previously unselected package libgfortran5:amd64.
Preparing to unpack .../04-libgfortran5_14.2.0-4ubuntu2~24.04.1_amd64.deb ...
Unpacking libgfortran5:amd64 (14.2.0-4ubuntu2~24.04.1) ...
Selecting previously unselected package libgfortran-13-dev:amd64.
Preparing to unpack .../05-libgfortran-13-dev_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking libgfortran-13-dev:amd64 (13.3.0-6ubuntu2~24.04.1) ...
Selecting previously unselected package gfortran-13-x86-64-linux-gnu.
Preparing to unpack .../06-gfortran-13-x86-64-linux-gnu_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking gfortran-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04.1) ...
Selecting previously unselected package gfortran-13.
Preparing to unpack .../07-gfortran-13_13.3.0-6ubuntu2~24.04.1_amd64.deb ...
Unpacking gfortran-13 (13.3.0-6ubuntu2~24.04.1) ...
Selecting previously unselected package gfortran-x86-64-linux-gnu.
Preparing to unpack .../08-gfortran-x86-64-linux-gnu_4%3a13.2.0-7ubuntu1_amd64.deb ...
Unpacking gfortran-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Selecting previously unselected package gfortran.
Preparing to unpack .../09-gfortran_4%3a13.2.0-7ubuntu1_amd64.deb ...
Unpacking gfortran (4:13.2.0-7ubuntu1) ...
Selecting previously unselected package libllvm17t64:amd64.
Preparing to unpack .../10-libllvm17t64_1%3a17.0.6-9ubuntu1_amd64.deb ...
Unpacking libllvm17t64:amd64 (1:17.0.6-9ubuntu1) ...
Selecting previously unselected package libamd-comgr2:amd64.
Preparing to unpack .../11-libamd-comgr2_6.0+git20231212.4510c28+dfsg-3build2_amd64.deb ...
Unpacking libamd-comgr2:amd64 (6.0+git20231212.4510c28+dfsg-3build2) ...
Selecting previously unselected package libhsakmt1:amd64.
Preparing to unpack .../12-libhsakmt1_5.7.0-1build1_amd64.deb ...
Unpacking libhsakmt1:amd64 (5.7.0-1build1) ...
Selecting previously unselected package libhsa-runtime64-1.
Preparing to unpack .../13-libhsa-runtime64-1_5.7.1-2build1_amd64.deb ...
Unpacking libhsa-runtime64-1 (5.7.1-2build1) ...
Selecting previously unselected package libamdhip64-5.
Preparing to unpack .../14-libamdhip64-5_5.7.1-3_amd64.deb ...
Unpacking libamdhip64-5 (5.7.1-3) ...
Selecting previously unselected package libevent-pthreads-2.1-7t64:amd64.
Preparing to unpack .../15-libevent-pthreads-2.1-7t64_2.1.12-stable-9ubuntu2_amd64.deb ...
Unpacking libevent-pthreads-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...
Selecting previously unselected package libhwloc15:amd64.
```

```
shylasri@vasudevkazipeta: ~
Selecting previously unselected package libhwloc15:amd64.
Preparing to unpack .../16-libhwloc15_2.10.0-1build1_amd64.deb ...
Unpacking libhwloc15:amd64 (2.10.0-1build1) ...
Selecting previously unselected package libmunge2:amd64.
Preparing to unpack .../17-libmunge2_0.5.15-4ubuntu0.1_amd64.deb ...
Unpacking libmunge2:amd64 (0.5.15-4ubuntu0.1) ...
Selecting previously unselected package libxnvctrl0:amd64.
Preparing to unpack .../18-libxnvctrl0_510.47.03-0ubuntu4.24.04.1_amd64.deb ...
Unpacking libxnvctrl0:amd64 (510.47.03-0ubuntu4.24.04.1) ...
Selecting previously unselected package ocl-icd-libopencl1:amd64.
Preparing to unpack .../19-ocl-icd-libopencl1_2.3.2-1build1_amd64.deb ...
Unpacking ocl-icd-libopencl1:amd64 (2.3.2-1build1) ...
Selecting previously unselected package libhwloc-plugins:amd64.
Preparing to unpack .../20-libhwloc-plugins_2.10.0-1build1_amd64.deb ...
Unpacking libhwloc-plugins:amd64 (2.10.0-1build1) ...
Selecting previously unselected package libpmix2t64:amd64.
Preparing to unpack .../21-libpmix2t64_5.0.1-4.1build1_amd64.deb ...
Unpacking libpmix2t64:amd64 (5.0.1-4.1build1) ...
Selecting previously unselected package librdmacm1t64:amd64.
Preparing to unpack .../22-librdmacm1t64_50.0-2ubuntu0.2_amd64.deb ...
Unpacking librdmacm1t64:amd64 (50.0-2ubuntu0.2) ...
Selecting previously unselected package libslurm40t64.
Preparing to unpack .../23-libslurm40t64_23.11.4-1.2ubuntu5_amd64.deb ...
Unpacking libslurm40t64 (23.11.4-1.2ubuntu5) ...
Selecting previously unselected package libucx0:amd64.
Preparing to unpack .../24-libucx0_1.16.0+ds-5ubuntu1_amd64.deb ...
Unpacking libucx0:amd64 (1.16.0+ds-5ubuntu1) ...
Selecting previously unselected package hwloc-nox.
Preparing to unpack .../25-hwloc-nox_2.10.0-1build1_amd64.deb ...
Unpacking hwloc-nox (2.10.0-1build1) ...
Selecting previously unselected package libmpich12:amd64.
Preparing to unpack .../26-libmpich12_4.2.0-5build3_amd64.deb ...
Unpacking libmpich12:amd64 (4.2.0-5build3) ...
Selecting previously unselected package mpich.
Preparing to unpack .../27-mpich_4.2.0-5build3_amd64.deb ...
Unpacking mpich (4.2.0-5build3) ...
Selecting previously unselected package libmpich-dev:amd64.
Preparing to unpack .../28-libmpich-dev_4.2.0-5build3_amd64.deb ...
Unpacking libmpich-dev:amd64 (4.2.0-5build3) ...
Setting up libslurm40t64 (23.11.4-1.2ubuntu5) ...
Setting up libevent-pthreads-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...
Setting up libgomp1:amd64 (14.2.0-4ubuntu2~24.04.1) ...
Setting up libxnvctrl0:amd64 (510.47.03-0ubuntu4.24.04.1) ...
Setting up gcc-13-base:amd64 (13.3.0-6ubuntu2~24.04.1) ...
Setting up libmunge2:amd64 (0.5.15-4ubuntu0.1) ...
Setting up libllvm17t64:amd64 (1:17.0.6-9ubuntu1) ...
Setting up libquadmath0:amd64 (14.2.0-4ubuntu2~24.04.1) ...
Setting up libhwloc15:amd64 (2.10.0-1build1) ...
Setting up libatomic1:amd64 (14.2.0-4ubuntu2~24.04.1) ...
Setting up libgfortran5:amd64 (14.2.0-4ubuntu2~24.04.1) ...
Setting up libubsan1:amd64 (14.2.0-4ubuntu2~24.04.1) ...
```

```
shylasri@vasudevkazipeta: ~
After this operation, 57.7 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 m4 amd64 1.4.19-4build1 [244 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 autoconf all 2.71-3 [339 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 autotools-dev all 20220109.1 [44.9 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 automake all 1:1.16.5-1.3ubuntu1 [558 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 javascript-common all 11+nmu1 [5936 B]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpsm-infinipath1 amd64 3.3+20.604758e7-6.3build1 [178 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpsm2-2 amd64 11.2.185-2build1 [194 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 libfabric1 amd64 1.17.0-3build2 [657 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 libopenmpi3t64 amd64 4.1.6-7ubuntu2 [2563 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcaf-openmpi-3t64 amd64 2.10.2+ds-2.1build2 [39.1 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcoarrays-dev amd64 2.10.2+ds-2.1build2 [37.5 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 openmpi-common all 4.1.6-7ubuntu2 [170 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/universe amd64 openmpi-bin amd64 4.1.6-7ubuntu2 [114 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcoarrays-openmpi-dev amd64 2.10.2+ds-2.1build2 [372 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [145 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-extra-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [64.2 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-openssl-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [15.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-dev amd64 2.1.12-stable-9ubuntu2 [274 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble/main amd64 libjs-jquery all 3.6.1+dfsg+~3.5.14-1 [328 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble/universe amd64 libjs-jquery-ui all 1.13.2+dfsg-1 [252 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble/main amd64 libltdl7 amd64 2.4.7-7build1 [40.3 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble/main amd64 libltdl-dev amd64 2.4.7-7build1 [168 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-3-dev amd64 3.7.0-0.3build1.1 [99.5 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-route-3-dev amd64 3.7.0-0.3build1.1 [216 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnuma-dev amd64 2.0.18-1ubuntu0.24.04.1 [37.0 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble/universe amd64 libhwloc-dev amd64 2.10.0-1build1 [268 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 zlib1g-dev amd64 1:1.3.dfsg-3.1ubuntu2.1 [894 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpmix-dev amd64 5.0.1-4.1build1 [4018 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble/main amd64 libtool all 2.4.7-7build1 [166 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libibverbs-dev amd64 50.0-2ubuntu0.2 [686 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble/universe amd64 libopenmpi-dev amd64 4.1.6-7ubuntu2 [864 kB]
Fetched 14.1 MB in 15s (908 kB/s)
Extracting templates from packages: 100%
Selecting previously unselected package m4.
(Reading database ... 47166 files and directories currently installed.)
Preparing to unpack .../00-m4_1.4.19-4build1_amd64.deb ...
Unpacking m4 (1.4.19-4build1) ...
Selecting previously unselected package autoconf.
Preparing to unpack .../01-autoconf_2.71-3_all.deb ...
Unpacking autoconf (2.71-3) ...
Selecting previously unselected package autotools-dev.
Preparing to unpack .../02-autotools-dev_20220109.1_all.deb ...
Unpacking autotools-dev (20220109.1) ...
Selecting previously unselected package automake.
Preparing to unpack .../03-automake_1%3a1.16.5-1.3ubuntu1_all.deb ...
Unpacking automake (1:1.16.5-1.3ubuntu1) ...
Selecting previously unselected package javascript-common.
Preparing to unpack .../04-javascript-common_11+nmu1_all.deb ...
Unpacking javascript-common (11+nmu1) ...
```

```
Preparing to unpack .../30-libopenmpi-dev_4.1.6-7ubuntu2_amd64.deb ...
Unpacking libopenmpi-dev:amd64 (4.1.6-7ubuntu2) ...
Setting up javascript-common (11+nmu1) ...
Setting up libcoarrays-dev:amd64 (2.10.2+ds-2.1build2) ...
Setting up libevent-openssl-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...
Setting up m4 (1.4.19-4build1) ...
Setting up libevent-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...
Setting up libnuma-dev:amd64 (2.0.18-1ubuntu0.24.04.1) ...
Setting up autotools-dev (20220109.1) ...
Setting up libltdl7:amd64 (2.4.7-7build1) ...
Setting up autoconf (2.71-3) ...
Setting up libevent-extra-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...
Setting up zlib1g-dev:amd64 (1:1.3.dfsg-3.1ubuntu2.1) ...
Setting up libnl-3-dev:amd64 (3.7.0-0.3build1.1) ...
Setting up libpsm2-2 (11.2.185-2build1) ...
Setting up openmpi-common (4.1.6-7ubuntu2) ...
Setting up libpsm-infinipath1 (3.3+20.604758e7-6.3build1) ...
update-alternatives: using /usr/lib/libpsm1/libpsm_infinipath.so.1.16 to provide /usr/lib/x86_64-linux-gnu/libpsm_infinipath.so.1 (libpsm_infinipath.so.1) in auto mode
Setting up libjs-jquery (3.6.1+dfsg+~3.5.14-1) ...
Setting up automake (1:1.16.5-1.3ubuntu1) ...
update-alternatives: using /usr/bin/automake-1.16 to provide /usr/bin/automake (automake) in auto mode
Setting up libfabric1:amd64 (1.17.0-3build2) ...
Setting up libtool (2.4.7-7build1) ...
Setting up libnl-route-3-dev:amd64 (3.7.0-0.3build1.1) ...
Setting up libltdl-dev:amd64 (2.4.7-7build1) ...
Setting up libjs-jquery-ui (1.13.2+dfsg-1) ...
Setting up libevent-dev (2.1.12-stable-9ubuntu2) ...
Setting up libopenmpi3t64:amd64 (4.1.6-7ubuntu2) ...
Setting up libhwloc-dev:amd64 (2.10.0-1build1) ...
Setting up libpmix-dev:amd64 (5.0.1-4.1build1) ...
Setting up openmpi-bin (4.1.6-7ubuntu2) ...
update-alternatives: using /usr/bin/mpirun.openmpi to provide /usr/bin/mpirun (mpirun) in auto mode
update-alternatives: using /usr/bin/mpicc.openmpi to provide /usr/bin/mpicc (mpi) in auto mode
Setting up libibverbs-dev:amd64 (50.0-2ubuntu0.2) ...
Setting up libcaf-openmpi-3t64:amd64 (2.10.2+ds-2.1build2) ...
Setting up libopenmpi-dev:amd64 (4.1.6-7ubuntu2) ...
update-alternatives: using /usr/lib/x86_64-linux-gnu/openmpi/include to provide /usr/include/x86_64-linux-gnu/mpi (mpi-x86_64-linux-gnu) in auto mode
Setting up libcoarrays-openmpi-dev:amd64 (2.10.2+ds-2.1build2) ...
update-alternatives: using /usr/lib/x86_64-linux-gnu/open-coarrays/openmpi/bin/caf to provide /usr/bin/caf.openmpi (caf-openmpi) in auto mode
update-alternatives: using /usr/bin/caf.openmpi to provide /usr/bin/caf (caf) in auto mode
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for man-db (2.12.0-4build2) ...
shylasri@vasudevkazipeta:~$
```

```
shylasri@vasudevkazipeta:~$ mpicc --version
mpirun --version
gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

mpirun (Open MPI) 4.1.6

Report bugs to http://www.open-mpi.org/community/help/
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```
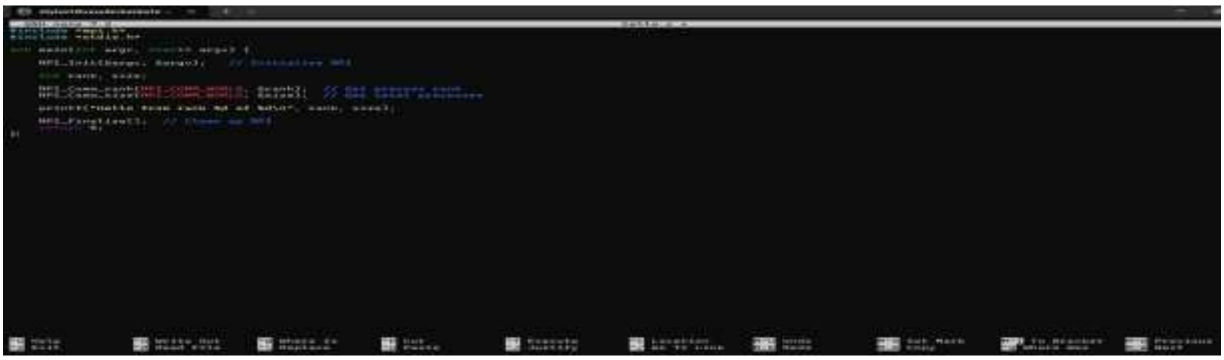

Task 1: MPI Hello World

Step 1: Create a file named hello.c with the following code:

```
#include &lt;mpi.h&gt;

#include &lt;stdio.h&gt;

int main(int argc, char** argv) {

MPI_Init(&amp;argc, &amp;argv);

// Initialize MPI

int rank, size;

MPI_Comm_rank(MPI_COMM_WORLD, &amp;rank);

// Get process rank

MPI_Comm_size(MPI_COMM_WORLD, &amp;size);

// Get total number of processes

printf(&quot;Hello from rank %d of %d\n&quot;, rank,

size);

MPI_Finalize(); // Clean up MPI

return 0;

}
```

Code Compilation Screenshot



Step 1:

Create a file named reduction_bad.c

Implement parallel summation using:

#pragma omp critical

inside the loop.

Screenshot

```
shylasri@vasudevkazipeta:~$ nano reduction_bad.c
```

Compile:

gcc -O2 -fopenmp reduction_bad.c -o reduction_bad

Screenshot

```
shylasri@vasudevkazipeta:~$ nano reduction_bad.c
shylasri@vasudevkazipeta:~$ gcc -O2 -fopenmp reduction_bad.c -o reduction_bad
```

Run with Thread 1:

export OMP_NUM_THREADS=1

Screenshot

```
shylasri@vasudevkazipeta:~$ nano reduction_bad.c
shylasri@vasudevkazipeta:~$ gcc -O2 -fopenmp reduction_bad.c -o reduction_bad
shylasri@vasudevkazipeta:~$ export OMP_NUM_THREADS=1
shylasri@vasudevkazipeta:~$ ./reduction_bad
Sum = 4999999950000000
Execution Time = 3.013773 seconds
```

Run with Thread 4:

export OMP_NUM_THREADS=4

Record execution time.

Screenshot

```
shylasri@vasudevkazipeta:~$ export OMP_NUM_THREADS=4
shylasri@vasudevkazipeta:~$ ./reduction_bad
Sum = 4999999950000000
Execution Time = 15.176758 seconds
```

Step 2: Compile the program using mpicc:

> mpicc -o hello hello.c

Screenshot

```
shylasri@vasudevkazipeta:~$ mpicc -o hello hello.c
```

Step 3: Run the program with different numbers of processes (e.g.,

2, 4) and observe the output:

mpirun -np 2 ./hello

mpirun -np 4 ./hello

Run with Different Number of Processes

Run with 2 Processes

Screenshot

```
shylasri@vasudevkazipeta:~$ mpirun -np 2 ./hello
Hello from rank 1 of 2
Hello from rank 0 of 2
```

```
shylasri@vasudevkazipeta:~$ mpicc -o hello hello.c
shylasri@vasudevkazipeta:~$ Hello from rank 0 of 2
Hello from rank 1 of 2
Command 'Hello' not found, did you mean:
  command 'hello' from snap hello (2.10)
  command 'hello' from deb hello (2.10-3)
  command 'hello' from deb hello-traditional (2.10-6)
  command 'jello' from deb jello (1.6.0-1)
See 'snap info <snapname>' for additional versions.
Command 'Hello' not found, did you mean:
  command 'hello' from snap hello (2.10)
  command 'jello' from deb jello (1.6.0-1)
  command 'hello' from deb hello (2.10-3)
  command 'hello' from deb hello-traditional (2.10-6)
See 'snap info <snapname>' for additional versions.
```

Run with 4 Processes

```
shylasri@vasudevkazipeta:~$ mpirun -np 4 ./hello
Hello from rank 0 of 4
Hello from rank 1 of 4
Hello from rank 2 of 4
Hello from rank 3 of 4
```

```
shylasri@vasudevkazipeta:~$ mpicc --version
mpirun --version
gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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mpirun (Open MPI) 4.1.6

Report bugs to http://www.open-mpi.org/community/help/
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 4: Answer the following questions:

1. What do MPI_Init and MPI_Finalize do?

MPI_Init()

- Initializes the MPI execution environment.
- Must be called before any other MPI function.
- Sets up communication between processes.

MPI_Finalize()

- Terminates the MPI environment.
- Cleans up all MPI resources.
- Must be the last MPI function called.

2. What is the range of ranks for a given number of processes?

If total number of processes = N

0 to N – 1

Example:

- If N = 2 → ranks are 0, 1
- If N = 4 → ranks are 0, 1, 2, 3

Each process has a unique rank.

3. Why can the output order be different each time you run?

To analyze synchronization overhead and load imbalance in

OpenMP programs and optimize performance using proper

directives.

Because:

- MPI processes run in parallel.
- The operating system scheduler decides execution order.
- Processes may finish at different times.
- There is no synchronization in printing.

Therefore, the output order is non-deterministic.

Observation

When the MPI program was executed with multiple processes, each process printed its rank and total number of processes. The output order changed in different runs.

In the OpenMP program, execution time increased when using more threads due to synchronization overhead.

Explanation

In MPI, each process runs independently and is assigned a unique rank from 0 to N−1. Since processes execute in parallel, the print order is not fixed.

In OpenMP, using a critical section forces threads to wait for each other, causing synchronization overhead and reduced performance when the number of threads increases.

Conclusion

This experiment helped in understanding:

- Basic parallel process execution using MPI.
- Rank assignment and communication setup in MPI.
- Synchronization overhead in OpenMP.
- How improper synchronization can reduce performance instead of improving it.
- The importance of choosing appropriate parallel directives for optimization.

Task 2: Simple Send/Receive – Oneway Communication

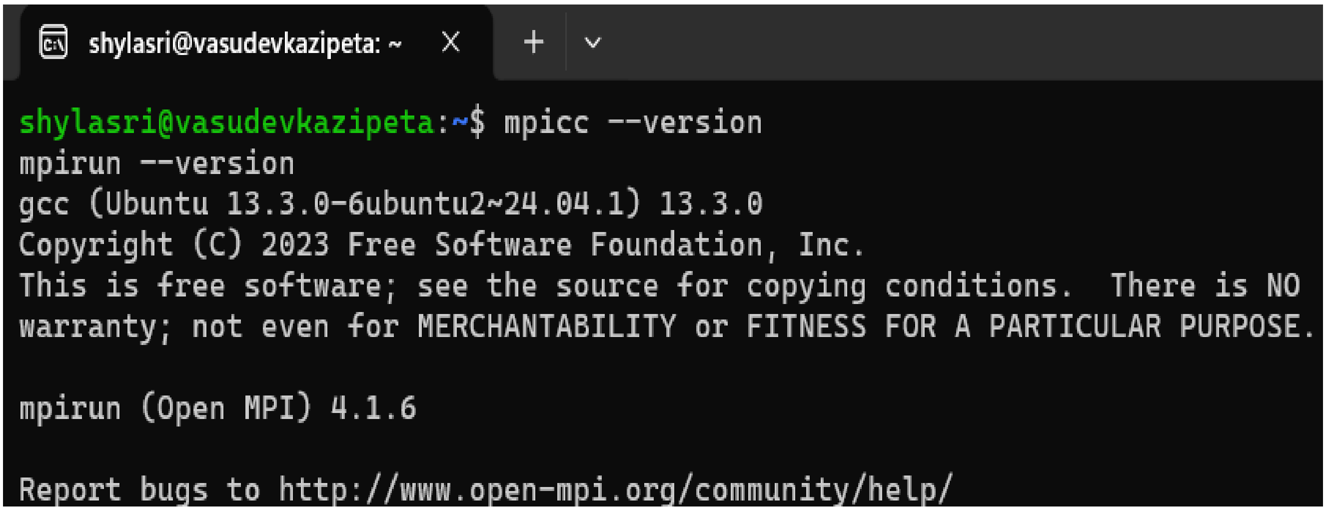
Step 1: Create a file named ping.c that sends an integer from rank

0 to rank 1.

```
#include &lt;mpi.h&gt;

#include &lt;stdio.h&gt;

int main(int argc, char** argv) {

MPI_Init(&amp;argc, &amp;argv);

int rank;

MPI_Comm_rank(MPI_COMM_WORLD, &amp;rank);

int data;
```

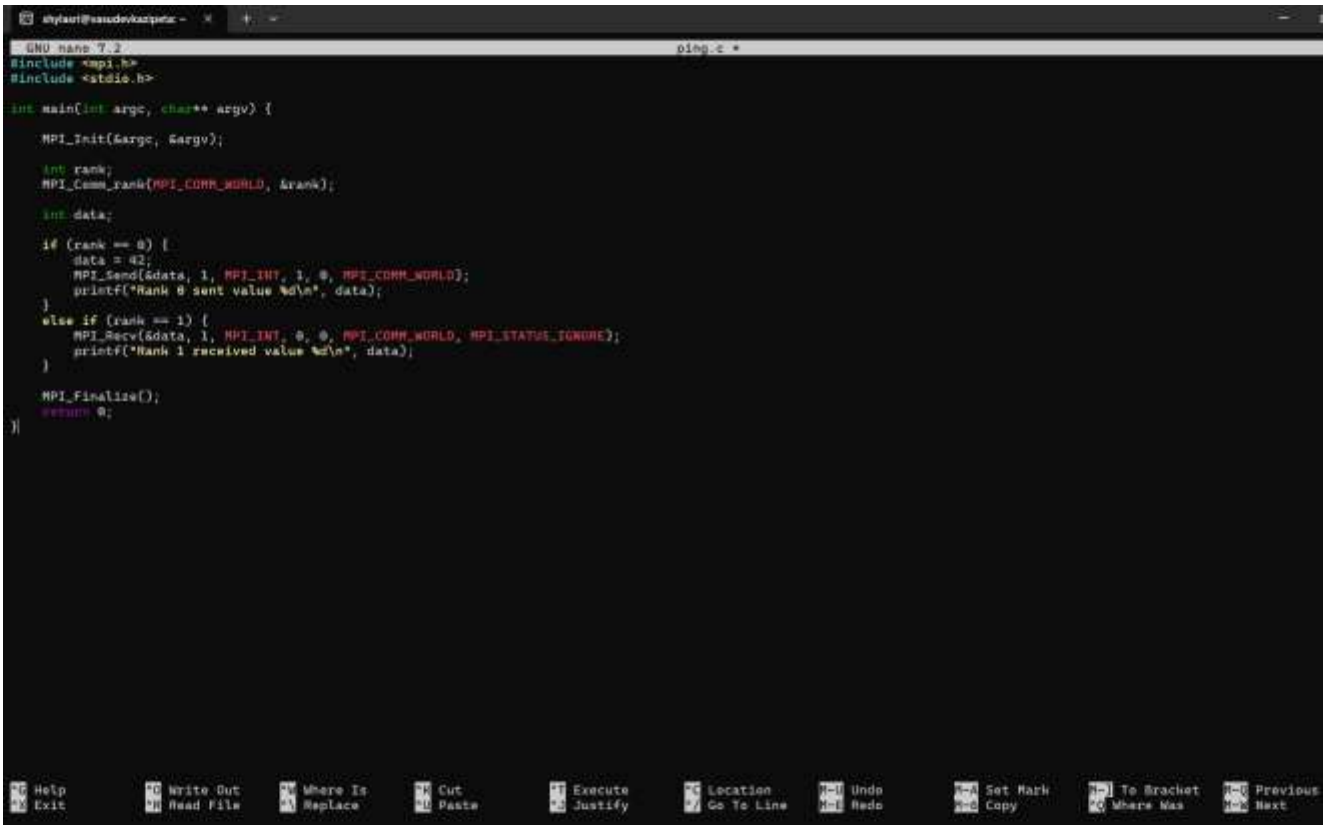
```
if (rank == 0) {

data = 42;

MPI_Send(&amp;data, 1, MPI_INT, 1, 0,

MPI_COMM_WORLD);

printf(&quot;Rank 0 sent value %d\n&quot;, data);

} else if (rank == 1) {

MPI_Recv(&amp;data, 1, MPI_INT, 0, 0,

MPI_COMM_WORLD, MPI_STATUS_IGNORE);

printf(&quot;Rank 1 received value %d\n&quot;, data);

}

MPI_Finalize();

return 0;

}
```

Screenshot



STEP 1: Create ping.c

Screenshot

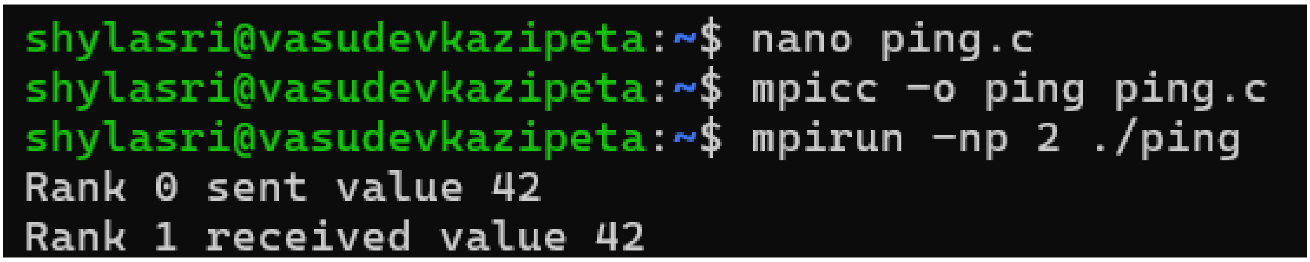


Step 2: Compile and run with exactly 2 processes:

mpicc -o ping ping.c

mpirun -np 2 ./ping

Screenshot



Step 3: Modify the program so that rank 1 sends a different value (e.g., 99) back to rank 0, and rank 0 receives it. This is a two-way communication (ping-pong). Write the code in pingpong.c and test it.

Two-Way Communication (Ping-Pong)

Step 1: Create pingpong.c

```
GNU nano 7.2                                   pingpong.c *
#include <mpi.h>
#include <stdio.h>

int main(int argc, char** argv) {

    MPI_Init(&argc, &argv);

    int rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    int data;

    if (rank == 0) {
        data = 42;
        MPI_Send(&data, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
        printf("Rank 0 sent value %d\n", data);

        MPI_Recv(&data, 1, MPI_INT, 1, 1, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        printf("Rank 0 received value %d\n", data);
    }
    else if (rank == 1) {
        MPI_Recv(&data, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        printf("Rank 1 received value %d\n", data);

        data = 99;
        MPI_Send(&data, 1, MPI_INT, 0, 1, MPI_COMM_WORLD);
        printf("Rank 1 sent value %d\n", data);
    }

    MPI_Finalize();
    return 0;
}

^G Help      ^O Write Out  ^W Where Is  ^K Cut    ^T Execute  ^C Location  M-U Undo  M-A Set Mark  M-] To Bracket  M-Q Previous
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy  ^Q Where Was  M-W Next
```

```
shylasri@vasudevkazipeta:~$ nano pingpong.c
```

Step 2: Compile

```
shylasri@vasudevkazipeta:~$ mpicc -o pingpong pingpong.c
```

Step 3:

```
shylasri@vasudevkazipeta:~$ mpirun -np 2 ./pingpong
Rank 0 sent value 42
Rank 0 received value 99
Rank 1 received value 42
Rank 1 sent value 99
```

Step 4: Answer the following:

1. What would happen if you run the program with more than 2 processes? Why?

If the program is run with more than 2 processes, only rank 0 and rank 1 will perform send and receive operations. The additional processes (rank 2, 3, etc.) will not participate in communication and will simply initialize and finalize MPI without doing anything.

The program is designed specifically for 2 processes, so using more processes may lead to idle processes or unexpected behavior.

2. What is the purpose of the tag parameter in MPI_Send and MPI_Recv?

The tag parameter is used to identify and match messages between sender and receiver.

It helps distinguish between different types of messages and ensures that the correct message is received. The sender and receiver must use the same tag value for successful communication.

3. What does MPI_STATUS_IGNORE do?

MPI_STATUS_IGNORE is used when the program does not need status information from MPI_Recv.

Normally, MPI provides details like the source, tag, and error status of the received message. If this information is not required, MPI_STATUS_IGNORE tells MPI to ignore it.

Observation

When the program was executed with 2 processes:

- Rank 0 successfully sent an integer value.
- Rank 1 successfully received the value.
- In the ping-pong program, rank 1 sent a different value back to rank 0.
- The communication occurred correctly between the two processes.

The output order sometimes changed because both processes execute in parallel.

Explanation

This experiment demonstrates basic message passing using MPI.

- MPI_Send() is used to send data from one process to another.
- MPI_Recv() is used to receive data.
- Communication occurs between processes identified by their ranks.
- The tag parameter ensures proper matching of send and receive operations.
- In two-way communication (ping-pong), both processes act as sender and receiver.
- If the order of send/receive is incorrect, the program may hang due to waiting.

This shows how MPI enables process-to-process communication in parallel programs.

Conclusion

This experiment helped in understanding:

- Basic one-way communication using MPI.
- Two-way (ping-pong) communication between processes.
- The importance of matching send and receive operations.
- The role of ranks and tags in MPI communication.

It demonstrates fundamental concepts of message passing in parallel computing.

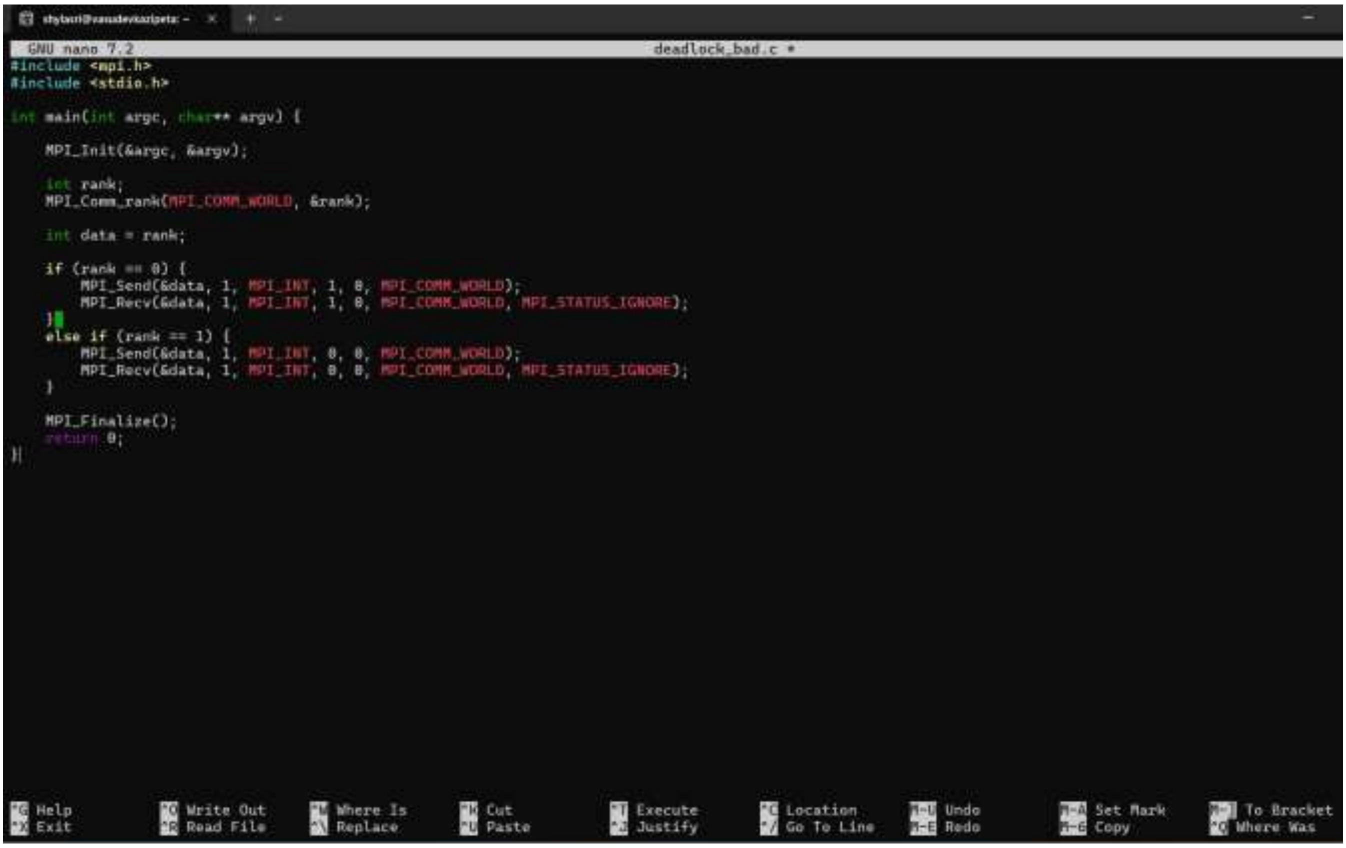
Task 3: Deadlock and Avoidance

Step 1: Create a file named deadlock_bad.c that attempts a circular send/receive:

```
#include &lt;mpi.h&gt;
#include &lt;stdio.h&gt;
int main(int argc, char** argv) {
MPI_Init(&amp;argc, &amp;argv);
int rank;
MPI_Comm_rank(MPI_COMM_WORLD, &amp;rank);
int data = rank;
if (rank == 0) {
MPI_Send(&amp;data, 1, MPI_INT, 1, 0,
MPI_COMM_WORLD);
MPI_Recv(&amp;data, 1, MPI_INT, 1, 0,
MPI_COMM_WORLD, MPI_STATUS_IGNORE);
} else if (rank == 1) {
MPI_Send(&amp;data, 1, MPI_INT, 0, 0,
MPI_COMM_WORLD);
MPI_Recv(&amp;data, 1, MPI_INT, 0, 0,
MPI_COMM_WORLD, MPI_STATUS_IGNORE);
}
MPI_Finalize();
return 0;
}
```

STEP 1: Create deadlock_bad.c

Screenshot



Step 2: Compile and run with 2 processes. Observe what happens

(the program will hang – you may need to kill it with Ctrl+C).

Compile

Screenshot

2 Processes

Screenshot

What Happens?

- Program will **hang**
- No output appears
- It does not terminate

Step 3: Now create a fixed version deadlock_fixed.c by reordering the calls so that one process receives first:

```
#include &lt;mpi.h&gt;
#include &lt;stdio.h&gt;
int main(int argc, char** argv) {
MPI_Init(&amp;argc, &amp;argv);
int rank;
MPI_Comm_rank(MPI_COMM_WORLD, &amp;rank);
int data = rank;
if (rank == 0) {
MPI_Send(&amp;data, 1, MPI_INT, 1, 0,
MPI_COMM_WORLD);
MPI_Recv(&amp;data, 1, MPI_INT, 1, 0,
MPI_COMM_WORLD, MPI_STATUS_IGNORE);
} else if (rank == 1) {
MPI_Recv(&amp;data, 1, MPI_INT, 0, 0,
MPI_COMM_WORLD, MPI_STATUS_IGNORE);
MPI_Send(&amp;data, 1, MPI_INT, 0, 0,
MPI_COMM_WORLD);
}
```

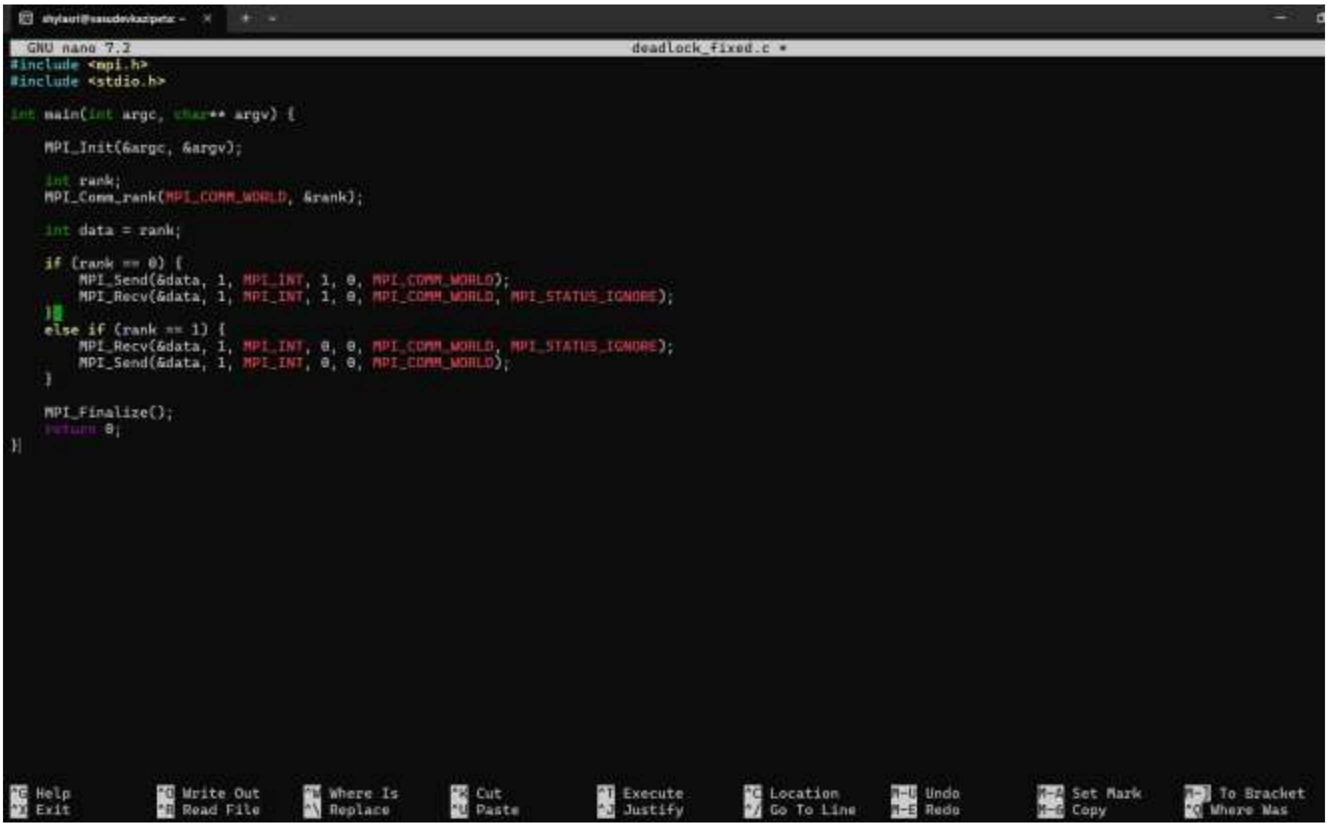
MPI_Finalize();

return 0;

}

STEP 3: Create Fixed Version deadlock_fixed.c

Screenshot



Step 4: Compile and run the fixed version. It should complete without hanging.

Screenshots

What Happens?

- Program runs successfully.
- No hanging.
- Execution completes normally.

Step 5: Answer the following questions:

1. Why does the first version deadlock? Explain in terms of the order of calls.

In the first version:

- Process 0 calls MPI_Send() first.
- Process 1 also calls MPI_Send() first.
- Both processes attempt to send data at the same time.
- MPI_Send() is a blocking call.
- A blocking send may wait until a matching receive is posted.

Since:

- Process 0 is waiting for Process 1 to receive.
- Process 1 is waiting for Process 0 to receive.

Neither process reaches the MPI_Recv() statement.

This creates:

- Circular waiting condition
- No progress
- Infinite blocking

This situation satisfies one of the main conditions of deadlock:

Each process is waiting for a resource held by the other.

Therefore, the program hangs indefinitely.

2. How does the fixed version avoid deadlock?

In the fixed version:

- Process 0 executes MPI_Send() first.
- Process 1 executes MPI_Recv() first.

Now the sequence becomes:

1. Process 0 sends data.
2. Process 1 is already ready to receive.
3. Communication completes successfully.
4. Process 1 then sends.
5. Process 0 receives.

Since one process is receiving while the other is sending:

- There is no circular waiting.
- No process blocks indefinitely.
- Communication proceeds smoothly.

Thus, proper ordering removes the deadlock.

3. Could buffering ever prevent this deadlock? Why is it

unsafe to rely on buffering?

Sometimes, MPI implementations buffer small messages internally.

If buffering occurs:

- MPI_Send() may copy data into buffer.
- The send call may return immediately.
- The program might not deadlock.

However, relying on buffering is unsafe because:

- Buffer size is limited.

- Large messages may not be buffered.
- Behavior depends on MPI implementation.
- Different systems may behave differently.

Therefore:

Deadlock may still occur in many cases.

Correct program design should not depend on buffering.

Observation

- The program deadlock_bad.c hangs during execution.
- Both processes wait indefinitely.
- The program deadlock_fixed.c executes successfully without hanging.
- Proper ordering of communication prevents deadlock.

Explanation

Deadlock occurs when two processes wait for each other to complete an operation.
In the bad version, both processes try to send first, leading to circular waiting.
In the fixed version, one process receives first, breaking the circular wait and allowing successful communication.

Conclusion

This experiment demonstrated how improper ordering of blocking MPI communication causes deadlock.
By reordering send and receive operations, the deadlock was avoided.
Careful communication design is essential in parallel programming.